

Systematic Green Design in Production Equipment Investments: Conceptual Development and Outlook

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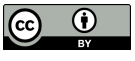
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Systematic Green Design in Production Equipment Investments: Conceptual Development and Outlook

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Abstract. This paper explores the concept of green design in the context of production, focusing on investment projects for production equipment design and acquisition by a manufacturing firm. Research towards making manufacturing and production related activities more sustainable is increasing. In the manufacturing sector, environmental sustainability tends to be more commonly approached from the operations perspective. However, the decisions taken in the design phase of the production equipment significantly impact the operations phase. Therefore, proactive design approaches for sustainability applied in product design settings could be transferred to the design of the production equipment to build in green aspects from the outset. This study explores the research questions of what green production equipment design entails and how the concept of green design has evolved in the context of production. Overall, this conceptual paper highlights the importance of incorporating green design principles from the outset of the production design. Transferable methodological issues are also explored for further detailed investigation in the production equipment design context. Strong collaboration between equipment suppliers and the buying manufacturer that aims to integrate sustainability as part of requirements is proposed as an enabler for the way forward. The paper also provides insights into the evolution of the concept in this context for possible future research.

Keywords: Production equipment · Design for sustainability · Green design

1 Introduction

As knowledge about the impact of industrial activities on the planet improves, the breadth of the research discourse on sustainability and green considerations in manufacturing and production also increases [1, 2]. This development could be viewed concerning

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the integration of conceptual elements, theoretical frames and perspectives adopted, methodologies and tools applied, and the broadening scope of empirical research and industrial application.

In the manufacturing context, environmental sustainability (commonly referred as green) is often approached from the operations execution perspective rather than from the design (or development) perspective [3]. However, it is well known that decisions taken in the design phase greatly impact the operations phase. First, studies have established that broad-based proactive approaches aiming at different activities from design to manufacturing to managing supply chains are important for achieving tangible sustainability gains [4]. Second, decisions made during the design phase are found to limit most alternatives applicable in subsequent stages. These proactive approaches frequently applied in product design could hence be transferred to the design of the production system to build green aspects from the outset.

The production system is generally highly complex, comprised of facilities, equipment and machines, people, material, information, logistics, and management systems to organize and manage the production system. Although design for X approaches could be applied to the production system broadly [5], they suit the investments in production equipment and machines very well since they are often one-piece products or interconnections of several stand-alone machines. Consequently, a green approach to production equipment design could be a promising way to reduce the environmental impact of the production system during not only the operations phase but during its whole life cycle, and the life cycle of items produced using the manufacturing equipment.

Many manufacturers engage in a continued effort of designing and acquiring production equipment, which involves multi-competence team organization for large manufacturers, and is very resource intensive. Growing interest has been noted recently in leveraging environmental impact reduction potentials through managing the production equipment design process. This could be reasoned out from two perspectives. Fulfilling sustainability goals requires systems thinking; in manufacturing, one needs to consider not only the design of the product but also the production equipment from the early design phases. Large manufacturers also continuously engage in these high-investment projects. Lack of a systematic means to bring sustainability as part of equipment design requirement makes green design in production a critical area that must be investigated.

Considerations regarding the green design approach on production equipment and systems investments could be viewed in terms of product features, material use, and energy consumption issues as generic areas of interest within the green design. However, parameters of interest within each of these and their potential interdependencies are complex. Early considerations on green design often focused on waste prevention and (hazardous) material avoidance and reduction in fulfilling required features/functions of a product [6, 7] which apply to a broader context for even better value.

Gong et al., [8] argue that green design should pursue maximum economic efficiency and minimum environmental impact. Green design is a broad (and, to some extent, contextually dependent) term that needs to be defined and described further. What does green design for sustainability mean? Does it include the broad concept of circular economy in production? Researchers have developed tools for green design of various products, including complex manufactured products [9]. A crucial stage in green product

design is understanding how the end user uses it. In production equipment design, one needs to understand how the buying manufacturer uses the production equipment. Such integration includes aligning important and pre-established ways of working, such as green design integration to lean product design and production [9] for a holistic green manufacturing system [10].

This conceptual paper explores the tenets of green design in the context of production, particularly production equipment investment. The study sets out to answer the following research questions:

RQ1-What is green production equipment design?
RQ2- How has the concept of green design evolved in the context of production, and what lessons can we learn from this evolution?

2 Green Design for Better Sustainability in Production Context

2.1 What is Green Design?

In general, the term green design has been applied in product development, especially for consumer products [7] while to a lesser extent to the design/development of production equipment and machining tools as part of the overall production system design process. When applied to mechanical equipment (hardware), green design implies considerations of environmental friendliness during the design phase. However, as explained later, green design must be extended beyond the design phase to include the development process to volume production. As many other research publications do, the description assumes that being environmentally friendly is a feature that different stakeholders, including the end user, appreciate. From the description, extrapolations can be made regarding what needs to be included to ensure environmental friendliness and economic efficiency. It could imply that some sustainable production principles and practices must be followed [11]. Here manufacturing and production are used interchangeably.

To establish a common standpoint, we first look at different definitions of concepts in the vicinity of green design and design for sustainability. One of the older definitions of green design comes from OTA [6]: "a design process in which environmental attributes are treated as design objectives, rather than as constraints. A key point is that green design incorporates environmental objectives with minimum loss to product performance, useful life, or functionality". Dowie [12] states that green design entails design such that products can be disassembled and recycled. Similar ways of defining green design with a focus on the product can also be found also in recent papers, such as: "to make the products environment-friendly, safe to use and energy and resources saving" [13]. Similarly, sustainable manufacturing and green manufacturing are respectively defined as: "the actions, initiatives and techniques that positively affect the environmental, social or economic performance of a firm; helping to control or mitigate the impacts of the firm's operations in the triple bottom line" [11], and "the production processes that use input with relatively low environmental impacts" [13].

A relatively more comprehensive concept than green design is design for sustainability. Design for sustainability refers to the triple bottom line where social and economic needs are as important as the environmental dimensions. So, it can be understood as "to

address all dimensions of sustainability, looking at bigger systems and asking more fundamental questions about consumption and production" [14]. Design for sustainability can be defined as "technical and product-centric design towards a focus on large scale system-level changes, in which sustainability is understood as a socio-technical challenge" [1], and "a rational and structured process to create something new for solving sustainability-related problems" [15]. Circular product design strategies, often called R-strategies, are developed based on work from [16, 17]. Re-think strategy is often premiered as one of the top strategies of the resource hierarchy perspective and could be seen as a green design feature at a relevant system level.

Key elements of interest can be extracted from these and similar definitions regarding critical principles, strategies, and relevant approaches for green design (see Sect. 4.2).

2.2 Developments Around Green Design

According to extant literature [1], the early examples of green design in practice mainly focused on lowering environmental impact by redesigning individual product qualities. Environmental legislation, corporate image and public perception, consumer demand, and rising waste disposal costs pushed manufacturers towards green design in the '90s [12].

Companies with proactive strategies for sustainability have recognized that instilling structured ways of environmental impact improvement is needed to maintain better financial performances [18]. However, what can be included as environmental improvement action in manufacturing companies has broadened from time to time through understanding the cause-effect chains and interdependencies happening in industrial activities concerning different flows: material, energy, and information. For example, internal energy losses in manufacturing equipment might appear very small compared to the more visible losses due to planning or usage-related inefficiencies [19]. However, with a holistic view, considering a production line that the equipment is part of could reveal interdependencies that could exacerbate even external losses if not addressed at the root-at-design stage of equipment. These may even indicate the further potential of saving both energy (e.g., saving of hot exhaust gases) and material (e.g., choosing efficient equipment, or reducing chips and scraps as well as associated contaminations).

Historically, green design is a dominantly product-focused approach. Primarily, it makes sense to focus on the design stage of a consumer product to facilitate source reduction and management of waste along the different phases of the product life cycle, including considerations for remanufacturing, as recognized decades ago [20]. It is also a relatively more manageable way to justify the business case of environmentally oriented initiatives, such as redesigning functional aspects of a product. Improving manufacturing processes with a green focus has eventually caught up. The trend seems to have followed a move from a reactive approach (e.g., how can part of an existing process be marginally improved to save energy or wastewater) to a more proactive approach, e.g., how can manufacturing equipment be designed so that it has the features to save energy or other inputs in its operational life [21]. This indicates an evident expansion of the green design focus, from mainly focusing on the product to the production equipment design.

Green lean is another stream of literature heading for green production equipment or system design. Under this approach, the focus is given on integrating green thinking

into process efficiency - improving initiatives to a limited extent [9], and the design of production for an efficient energy use at a production facility level [19]. Built-in technologies for pollution control are becoming common practices in manufacturing equipment and process design. In addition, green-focused design considerations beyond the manufacturing shop floor could be found, extending to, for example, design for multi-use and modularity of packaging in manufacturing [22].

A challenge observed in extant literature from a conceptual point of view is that with these developments and various points of departure, the conceptual underpinnings of a holistic "sustainability-dominant logic" [23] are often undermined. Batista et al., [23] offer theoretical propositions for a sustainability-dominant logic for circular economy and sustainable operations at product, firm, and supply chain levels which could be used further to explore developments for green production equipment and system design. This could have implications for the circular economy for green design by enhancing an upstream and proactive focus rather than just downstream-focused strategies such as reuse, repurpose, and recycle [24]. As mentioned earlier, the development can be explained using sustainability-dominant logic at the product, manufacturing firm, and production value network levels. Essentially the transition seems to be towards behavioral change-based thinking in design for system innovations leading to more sustainable future manufacturing [1].

In their paper, Gaziulusoy et al., [25] present a scenario method linking product development to the visions of sustainability at the societal level. The aim is for companies to relate product development to the organization and society, compared to standard product-centered design and development approaches. According to the authors, the method allows for generating innovation paths for product development to respond to changes for sustainability at the system level.

Briefly, the answer to the question, what is green design? And what about its related concepts? It depends on the perspective, context, and overall societal changes and trends. The answer requires exploration across adjacent areas of application. That is why an explorative literature survey has been applied in this paper.

3 Methodology

This study employed a systematic literature review [26]. Steps from planning the review to conducting and presenting results have been followed as prescribed in [27]. The review started by picking a few seed publications based on the intended research questions and researchers' experience. These papers have been used to set search protocols and set initial keywords in the search query. Inclusions and exclusion criteria has been set as follows. Engineering, environmental studies, business, and management publications have been selected as a priority from subject areas. Based on the initial list, screening has been done to exclude medical and chemical subject areas. Only journal articles and book chapters written in English have been considered at this stage.

Initial Search Query Include from title, abstract and keywords: ("green design *" OR "green process design" OR "eco* design" OR "sustainable production principles" OR "environmental design" OR "design for sustainability").

Exclude from title, abstract and keywords: (*buildings* OR medic* OR bio* OR agricultur* OR crop* OR chemical* OR *buildings* OR urban OR architectu* OR "food production").

The literature search has been conducted using the Scopus® database. We executed the query in Web of Science® and the results were much fewer compared to the ones obtained via Scopus. Being aware of also the possible limitations and considering that Scopus indexes a slightly higher number of journals than Web of Science, we decided to continue only with Scopus.

The primary literature search focused on publications from the year 2000 to capture the most recent developments. According to Smart et al., [26], the output of the information search should be a full listing of articles and papers on which the review will be based. Skimming through titles and abstracts enabled us to shortlist for detailed review. In addition to the detailed review of the topically shortlisted papers, snowballing of essential papers, as suggested in Thome et al., [27], has been done to identify potentially relevant additional publications that were not included after the screening. Ultimately 69 titles from the main screening, and eleven titles from back- and forward tracking have resulted in 80 titles (as of 1 May 2023) for final review (see Fig. 1 for a summary of the screening process).

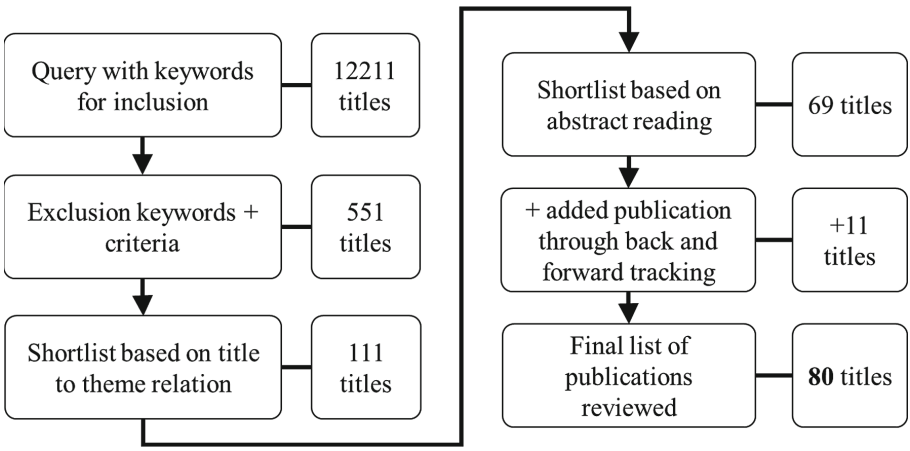


Fig. 1. The screening process for review

Figure 2 illustrates the trend of publications from the final shortlist covering the search period. The years that most reviewed titles were published in 2022 (14%), 2021 (10%), followed by 2017, 2013, and 2010 equally at 9%. Journal of Cleaner Production (50%) and Sustainability (Switzerland) (22%) journals contributed by far the highest proportion of papers included in the review.

Analytical exploration has been done based on three areas of interest, as seen in Sect. 4. The first is the exploration and identification of tenets of green design. The intent has been to observe developments and change patterns in the application context. The second relates to methodological issues relevant to the investigation and discussion of

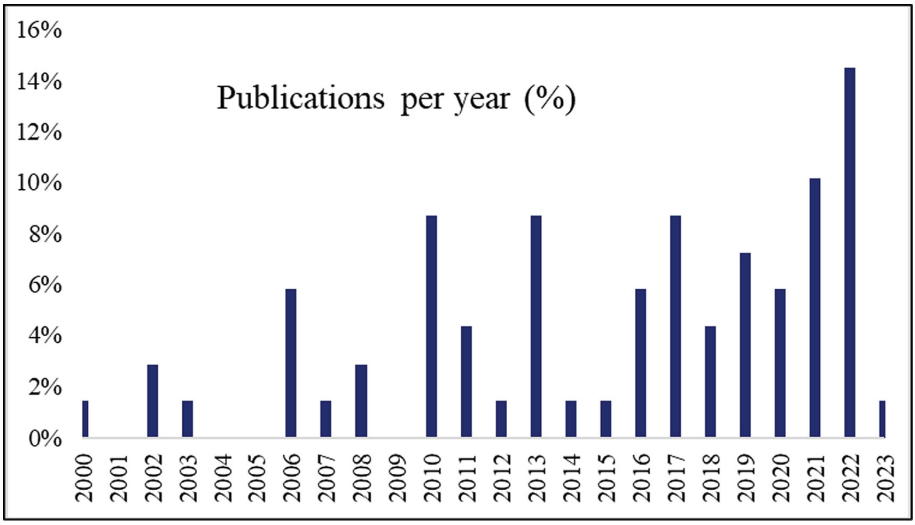


Fig. 2. Distribution of reviewed publications per year

green design research. The third is the exploration of theoretical underpinnings relevant to the green design discourse with the intent of eventually establishing conceptual framing to guide future research in systematic green design for production equipment as well as the broader capital investment for developing or updating production lines.

4 Findings from the Literature Review

4.1 Tenets of Green Design

Starting with the sample definitions presented in Sect. 2 and a preliminary review of the extant literature, one can observe that several aspects of green design become apparent. We consider asking these questions to support the exploration: (1) the bases of green design (what); (2) the purpose and motives of considering green design (why); and (3) methodological aspects: methods, approaches and tools of implementing different green design initiatives (how). The first two categories are presented in this section and in Table 1, while the how is described in Sect. 4.2 and in Table 2.

The reviewed literature indicates that guiding principles and corporate strategies should guide how this process is organized for optimal results. However, environmental objectives are not sufficient to keep businesses viable. They must be coupled with reasonable economic gain and not compromise social objectives. Streamlined green strategies can help an outward (market interest) focused linkage for green design initiatives [28, 29]. On the other hand, principles [11] are more inclined towards creating coherence among different elements of design-related processes in the organization that also could include decisions and relations with suppliers.

Principles and strategies for green design often manifest in practices employed or established processes (see Table 1). From the reviewed literature, it is apparent that green

design is often seen as a structured process in which environmental objectives must be an integral part of the deliverables of the design process [6, 13]. While a large portion of the literature analyzed views this process within the realm of designing products such as consumer goods [13], a large-scale system view of design consideration is becoming a recent trend.

Alternative terms such as design for sustainability [1] have been coined to reflect this broadening scope with a perspective that it should help address the underlying problem of un-sustainability [15]. Design for sustainability encompasses processes and sustainable design and production practices [30] related to, for example, eco-design, design for efficient machining, assembly and disassembly, green innovation, green manufacturing, and so on [3, 31]. Recent studies have emphasized the importance of organizational elements beyond purely technology-focused design thinking, such as information exchange, monitoring, and collaboration with suppliers (also known as green purchasing).

Table 1. Summary of key tenets in green design

What	Why
• Principles [11] • Strategies for green design: e.g., modular green design, eco-design [32] • Practices and people skills [30] • Established processes [9]	• Regulatory requirements: policies, local rules, global standards [33, 34] • Pressure from partners and stakeholders on environmental initiatives [35] • Performance improvement motives in multiple dimensions [28] • Company internal sustainability goals and pressure as drivers [36]

4.2 Methodological Aspects in Green Design Research

This section provides an inventory of methods and approaches used in green design as compiled from reviewed literature. Diverse research papers present a toolbox of different methods and approaches (see Table 2). In most of the reviewed papers discussing tools for green design, notions of "eco-design tools" has been dominant as a collective name for various tools used to capture especially environmental aspects in design. Under the umbrella of this notion, variants of life cycle (LCA), environmental benchmarking and several others have been mentioned. Several of the reviewed papers discussed life cycle costing (LCC), multi criteria analysis as well as analytical hierarchy theory, value stream mapping (VSM). Table 2 showcases some of the identified methods and tools for green design in the review, and related implications. Some methods and tools are data intensive and time-consuming (e.g., LCA, LCC), and others (VSM) are more customized and hands-on but still with their own limitations [3, 4].

The methods can be applied individually or in combination to alleviate the shortcomings and support different purposes within the design of products and production systems.

This approach invites hybridization [37], where it is relevant to analyze the design methods not as unified tunnels of the process but as toolboxes containing components that could be mixed and matched across methods.

Table 2. Green design methods and tools in the reviewed literature

Methods/tools	Purpose	Implication
• Industrial big data analytics [38] • Life cycle assessment [3] • IoT for energy efficiency [38] • Life cycle cost analysis [39] • Sustainable VSM, Environmental VSM [4]	• To design intelligent equipment • To estimate and improve the environmental impact of manufacturing • To visually assess manufacturing sustainability	• Using IoT to collect and analyze environmental data supporting sharing and collaboration with equipment designers/suppliers • Together with design for environment approach, it can be used for improving equipment design requirements • It allows to assess of the sustainability performance of manufacturing systems but requires developing a future state for improvement implementations

4.3 Theoretical Frames Accompanying Green Design Research

The dominant theoretical perspectives in green design research reviewed in this study include the dynamic capabilities view, institutional theory, resource dependency theory, and relational view. A recent study has proposed a "sustainability-dominant logic" [23] to theorize the phenomenon surrounding green design and circular economy.

From a dynamic capabilities perspective, integrating internal and external resources along with resource building and reconfiguration create dynamic capabilities with a stronger sustainability orientation [2]. In this framing, eco-design and green innovation foster market performance. Resource building and reconfiguration can be viewed in terms of hiring and developing environmental competencies, reconfiguring team arrangements in design projects, and improving processes for supplier relationships [28]. External resource integration can be interpreted to encompass orchestrating knowledge and competence resources from suppliers about materials, manufacturing process technologies, and environmental impact to embed better environmental performance as part of design requirements. While the technical competence of machine and equipment suppliers plays a vital role in designing more sustainable manufacturing facilities, mainly suppliers of production inputs for realizing the product (rather than the production process itself) have been emphasized.

In line with the external resource integration idea of the dynamic capabilities view, resource dependency theory posits that companies rely on resources in their business environment much more than the resources they possess, and essential resources are acquired through collaborations and enhanced relationships. This is partly because it is not feasible to own and have control over all strategic resources. An area of interest for environmental collaboration in manufacturing is, for example, environmental information sharing, education, and support to machine suppliers. Supplier selection based on environmental certification and fulfillment of environmental requirements in design specifications could be exploited via strong external collaborations [30]. Theoretical frames on intermediaries in sustainability transitions are relevant to understand the role of equipment suppliers and other actors as intermediaries and platforms [40]. These intermediaries can positively influence sustainability transition processes [41] by linking their related skills and resources or by connecting transformation, transition visions, and future demands to create momentum for socio-technical system change [40, 42].

Intangible resources revealed as the result of collaboration and negotiations are sources of coevolution of business actors in their sustainability effort [29]. Similar ideas of having long-lasting relationships based on trust and collaboration, even in competitive contexts, are addressed in the relational view theory [43].

5 Discussion and Future Outlook for Green Design in the Production Context

Design for sustainability has evolved from encompassing form and functional elements of a product in the early stages to include many socio-technical considerations and systems thinking to collaborative design thinking in recent developments [1, 15]. The application focus of a dominant sustainability logic has broadened to include levels from product to value creation/institution on the supply chain level, which seems consistent for design for sustainability and circular economy [23]. Building on the work of Batista et al., [23], we can argue, based on our conceptual exploration, that a sustainability-dominant logic seems to provide a coherent frame of reference for understanding the phenomenon of green and sustainable production equipment design in the broader sense than just a singular product item.

In its broader form, a sustainability-focused production equipment design encompasses collaborative approaches built on coherent strategies and principles for a synergistic environmental impact reduction from design (equipment and machines), manufacturing, and other support processes, including purchasing. In manufacturing equipment, the purchasing function is essential role in integrating of resources and capabilities from the manufacturer to those of the equipment suppliers involved in the investment projects, from design to commissioning to execution of the production facility. Challenges and knowledge or awareness asymmetries as to what is possible and required to convert sustainability ambitions into technical requirements and functional capabilities of new production equipment are inevitable. Working through close relationships to bring out higher levels of innovative knowledge towards a greener production equipment design shall be a way forward for manufacturers and equipment suppliers designing and commissioning the production equipment.

Further developing and clarifying the definition of green design and related terms such as circular design, green lean, sustainable design, etc., and their relation to the application area of how to design production equipment or even the whole production system for environmental sustainability and circularity is an interesting area for future research.

Expanding the requirements on existing production systems is forcing the industry to engage in also the renewal of their production systems. This renewal entails a shift in how production systems are designed and developed, including the physical setup of production equipment and machines and investments in infrastructure and facilities necessary to operate a factory. Integrating green targets as part of the requirements specification to be fulfilled in the designated production system is an important and practical way of managing the green design target. Material use, technology use for impact reduction for design and execution of the production system and change in awareness and competence development are needed for a more sustainable production environment. Collaboration among key value chain actors across both the product and production system lifecycle seems to be a significant evolutionary development of green production design [23].

Opportunities to keep production equipment efficient and environmentally sustainable – leading to the realization of greener products– are better exploited if sustainability has already been built into the production equipment. Small incremental changes in the operations phase will not influence the same level as will the initial design decisions.

Collaboration in the manufacturing value chain often focuses on function in the execution stages, in both cases, not necessarily on sustainable design [43]. To a considerable extent, social sustainability aspects related to ergonomics and accident reduction/avoidance in manufacturing environments are recognized in production design requirements. However, it seems only recently that environmental requirements are on the agenda for equipment purchase decisions.

Environmentally superior innovative ideas can be generated due to green design practices internal to the manufacturers and in interaction with knowledge and technology resources residing at the equipment supplier [2, 10, 15], which existing literature only rarely recognized.

The design principles, practices, tools and approaches, despite their differences in scope or applicability, imply more or less a consistent message. That is, pursuing sustainability on many fronts to improve environmental impact of production activities is relevant. If there are good and applicable practices in product design, we should enhance and use them at broader scale for production equipment design for even better environmental value [5]. At the same time, we should not forget about the social and economic aspects of sustainability but rather embed them as much as possible in the design for balancing gains. The target customer and context of use for production equipment are different from that of consumer products. This focus requires special care about which tools would be more suitable, and what standards and legal frameworks can positively influence the move towards green production equipment design. The EU directives for eco-design have been points of interest in some reviewed papers. There is an opportunity to explore recent developments in policy instruments (e.g., harmonized standards of the EU), specifically in equipment design for manufacturing.

6 Conclusion

This paper presented a conceptual exploitation of an emerging thought and discourse for green production equipment design. Some dominant theoretical underpinnings supporting conceptual development within green production design include systems view and relational views that describe the phenomenon of how closely the actors involved in the production equipment value chain (design, acquisition, operation) have to work to leverage their higher-order competencies and capabilities for a greener production equipment design and execution.

Both equipment suppliers and manufacturers buying and using the equipment need to engage in strong collaboration with green design as a top priority. Increasing awareness both at equipment supplier as well as manufacturer for green design appears an important element if collaboration for green design is to be effective.

The exploration of key tenets of green design and identification of applicable theoretical underpinnings is an important contribution of this study to push development of detailed frameworks and guidelines to advance green thinking as main constituent in production equipment design. Cross-disciplinary design events are examples that could be a point of future research interest in this direction.

The process of such guidelines development can be backed up by selective use of green design approaches. Different tools currently at use in product design have a potential to enable a more proactive application of principles and strategies for green design to a broader scope of green production equipment design.

The investigation in this study can be regarded as a preliminary exploration. As such, limitations in terms of identification and stipulation of some essential details will likely be rectified in a continued exploration. For instance, the systematic literature review could be expanded, including additional databases and potentially non-English literature.

Further research avenues include detailed conceptual analysis in line with a sustainability-dominant logic in green design and empirical exploration to further develop knowledge around the phenomenon [2, 23]. Together with conceptual developments, empirical investigation into the efforts of the collaborating actors is a vital element for expanding the discourse around green production equipment design.

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